

$$F_{\text{TCP}} = \begin{bmatrix} f \\ m \end{bmatrix} = \underbrace{A^\top \begin{bmatrix} K_p & \mathbf{0} \\ \mathbf{0} & K_R \end{bmatrix} A}_{K_{\text{TCP}}} \begin{bmatrix} e_p \\ e_R \end{bmatrix} + \underbrace{A^\top \begin{bmatrix} D_p & \mathbf{0} \\ \mathbf{0} & D_R \end{bmatrix} A}_{D_{\text{TCP}}} \begin{bmatrix} \dot{p}_d - \dot{p}_{\text{EE}} \\ \omega_d - \omega_{\text{EE}} \end{bmatrix}$$